

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	31/07/2026	Duration:	60 minutes

Code of Conduct

*I **K.Prathiba (192421163)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **K.Prathiba**

Annexure A – Agile Sprint Planning & User Story Workshop

AI-Based Personal Finance and Budget Planning System (Banking & Finance)

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders

1.1 Project Vision

- The vision of the **AI-Based Personal Finance and Budget Planning System** is to help individuals manage their personal finances efficiently using Artificial Intelligence.
- The system provides smart budgeting, expense tracking, income management, savings recommendations, bill reminders, and financial insights.
- By analyzing users' spending habits, the application offers personalized suggestions that encourage better financial planning and help users achieve their financial goals.

- The system aims to simplify financial management by providing an easy-to-use platform where users can monitor income, expenses, investments, and savings in one place. AI-based analysis predicts future expenses, identifies unnecessary spending, and recommends strategies to improve financial stability.

1.2 Project Objectives

The major objectives of the project are:

1. Develop an intelligent personal finance management system using Artificial Intelligence.
2. Allow users to record and categorize daily income and expenses.
3. Generate monthly and yearly budget plans automatically.
4. Provide AI-based recommendations to improve savings.
5. Predict future expenses based on historical financial data.
6. Send bill payment reminders to avoid late payment penalties.
7. Generate visual reports and financial dashboards.
8. Help users achieve financial goals through personalized planning.
9. Ensure secure storage of financial information.
10. Improve financial awareness and decision-making among users.

1.3 Project Scope

The proposed system covers various personal financial management activities, including:

- User registration and secure login
- Income management
- Expense tracking
- Budget creation
- Savings goal management
- AI-based financial recommendations
- Bill payment reminders
- Financial report generation
- Spending pattern analysis
- Data visualization using charts and graphs

The system is designed for students, employees, freelancers, and small business owners who wish to manage their finances effectively.

1.4 Key Stakeholders

Stakeholders are individuals or organizations involved in or affected by the project.

Stakeholder	Role in the Project
End User (Customer)	Uses the application to manage personal finances, track expenses, and receive AI-based recommendations.
Project Manager	Plans the project, manages schedules, resources, sprint activities, and ensures timely completion.
Business Analyst	Collects user requirements, analyzes business needs, and prepares software requirement specifications.
Product Owner	Defines product vision, prioritizes product backlog, and ensures the product meets customer expectations.
Scrum Master	Facilitates Agile practices, removes obstacles, and ensures smooth sprint execution.
Software Developers	Design, develop, integrate, and maintain the software modules.
UI/UX Designer	Creates user-friendly interfaces and improves user experience.
AI/ML Engineer	Develops AI algorithms for budgeting, spending prediction, and personalized recommendations.
Database Administrator (DBA)	Designs, manages, and secures the financial database.
Quality Assurance (QA) Tester	Tests the application, identifies bugs, and verifies software quality.
System Administrator	Maintains servers, deployment, backups, and system security.
Banking/Payment Service Providers	Provide secure payment gateway and transaction integration.
Government & Regulatory Authorities	Ensure compliance with financial regulations and data privacy laws.

1.5 Stakeholder Responsibilities

Stakeholder	Responsibilities
End User	Manage finances, provide feedback, and use system features.
Product Owner	Define product requirements and prioritize features.
Scrum Master	Conduct sprint planning, daily stand-ups, reviews, and retrospectives.
Project Manager	Monitor project progress and resource allocation.
Developers	Implement application features and fix defects.
AI Engineer	Build intelligent budgeting and prediction models.
UI/UX Designer	Design intuitive interfaces and dashboards.
QA Tester	Perform functional, integration, and system testing.
DBA	Ensure secure and efficient data storage.
System Administrator	Deploy and maintain the application infrastructure.

Summary

- The **AI-Based Personal Finance and Budget Planning System** aims to improve users' financial well-being through AI-driven budgeting, expense tracking, and personalized financial recommendations.

- A collaborative effort from stakeholders such as users, developers, project managers, AI engineers, and testers ensures the successful development and deployment of the system.

2. Create Epics and User Stories

Introduction

- The **AI-Based Personal Finance and Budget Planning System** is divided into multiple **Epics**, where each Epic represents a major feature of the application.
- Every Epic is further broken down into **User Stories** that describe the system requirements from the user's perspective.
- The user stories follow the Agile format: "**As a <user>, I want <feature>, so that <benefit>.**"
- This approach helps the development team understand user needs and deliver the project incrementally during each sprint.

Epics

- User Account Management
- Income & Expense Management
- Budget Planning
- AI-Based Financial Recommendations
- Savings Goal Management
- Reports & Dashboard
- Notifications & Bill Reminders
- Security & Data Management

Characteristics of Good User Stories

- User-focused
- Simple and understandable
- Independent

- Testable
- Valuable to customers

User Stories

- As a user, I want to register and log in securely, so that I can access my financial information safely.
- As a user, I want to add and manage my income, so that I can track my earnings accurately.
- As a user, I want to record and categorize my daily expenses, so that I can monitor my spending habits.
- As a user, I want to create monthly budgets, so that I can control my expenses effectively.
- As a user, I want AI to analyze my spending patterns, so that I can receive personalized financial recommendations.
- As a user, I want to set savings goals, so that I can achieve my future financial objectives.
- As a user, I want to view reports and graphical dashboards, so that I can easily understand my financial performance.
- As a user, I want to receive bill reminders and budget alerts, so that I can avoid late payments and overspending.

Benefits of Using User Stories

- Clearly captures customer requirements.
- Improves communication among Agile team members.
- Supports effective sprint planning.
- Helps prioritize product features.
- Enables continuous customer feedback.
- Reduces development risks.
- Improves software quality.
- Ensures customer satisfaction.

This format is suitable for a **Software Engineering Agile Sprint Planning** report and is detailed enough to score well in the assessment.

3. Define Acceptance Criteria

Introduction

- Acceptance Criteria define the conditions that must be satisfied before a user story is considered complete. They ensure that each feature works as expected and meets the user's requirements. In Agile, the **Given–When–Then** format is commonly used to describe testable and measurable conditions for each user story.

Acceptance Criteria (Given–When–Then)

1. User Registration

- **Given** a new user is on the registration page, **When** the user enters valid details and clicks **Register**, **Then** a new account should be created successfully.

2. Secure Login

- **Given** a registered user has valid credentials, **When** the user enters the correct email and password, **Then** the system should allow secure access to the dashboard.

3. Add Income

- **Given** the user is logged in, **When** the user enters income details and saves them, **Then** the income record should be stored and displayed successfully.

4. Record Expenses

- **Given** the user wants to track expenses, **When** expense details are entered and submitted, **Then** the system should save and categorize the expense correctly.

5. Create Budget

- **Given** the user wants to manage monthly spending, **When** a budget amount is entered and saved, **Then** the system should create the budget and display the remaining balance.

6. AI Financial Suggestions

- **Given** sufficient financial data is available, **When** the AI analyzes the user's spending habits, **Then** the system should provide personalized budgeting and savings recommendations.

7. View Reports

- **Given** the user has recorded financial transactions, **When** the user opens the Reports section, **Then** the system should display charts, summaries, and spending analysis accurately.

8. Bill Reminders

- **Given** a bill due date is stored in the system, **When** the due date approaches, **Then** the system should send a reminder notification to the user.

Summary

- These acceptance criteria provide clear, measurable, and testable conditions for the selected user stories.
- Using the **Given–When–Then** format ensures that every feature can be verified during testing and successfully accepted at the end of each Agile sprint.

4. Prioritize the Product Backlog

Introduction

- The **Product Backlog** is a prioritized list of all features and requirements for the **AI-Based Personal Finance and Budget Planning System**.
- It is organized based on **business value, customer needs, and project dependencies** to ensure that the most important features are developed first.
- Core functionalities required for the successful operation of the system are given higher priority, while additional enhancements are scheduled for later sprints.

High Priority

- User Registration and Login
- Income Management
- Expense Management
- Budget Planning

Medium Priority

- AI-Based Financial Recommendations
- Savings Goal Management
- Reports and Dashboard

Low Priority

- Bill Payment Reminders
- Notification Settings
- Profile Management
- User Preference Customization

Product Backlog Prioritization

- The highest priority is given to **User Registration and Login**, as users must create an account and securely access the system before using any other feature.
- After authentication, **Income Management** and **Expense Management** are prioritized because they allow users to record and manage their financial transactions.
- Once these core financial records are available, the **Budget Planning** module is implemented to help users create monthly budgets and monitor their spending.
- The next priority is the **AI-Based Financial Recommendation** feature, which analyzes users' financial data and provides personalized budgeting and savings suggestions.
- After implementing AI recommendations, the **Savings Goal Management** feature is developed to enable users to set and track their financial goals.
- The **Reports and Dashboard** module is then added to display financial summaries, charts, and spending analysis for better decision-making.
- Finally, supporting features such as **Bill Payment Reminders**, **Notification Settings**, and **Profile Management** are implemented.
- These features improve user convenience by sending timely alerts, allowing customization of notifications, and enabling users to update their personal information and account settings.

Summary

- Prioritizing the product backlog ensures that essential features with the highest business value are delivered first, while less critical features are implemented in later sprints.
- This approach supports Agile development by providing a functional system early, reducing project risks, and continuously delivering value to users.
- It improves collaboration among stakeholders and the development team by ensuring that everyone focuses on the most important project requirements during each sprint.
- Regular backlog prioritization also enables the team to adapt to changing customer needs, incorporate feedback, and enhance the overall quality and usability of the AI-Based Personal Finance and Budget Planning System.

5. Plan the Sprint Backlog

Introduction

The **Sprint Backlog** includes the highest-priority user stories selected from the product backlog for the first sprint. These user stories are divided into smaller development tasks to ensure that the team can complete them within the sprint duration. The primary goal of Sprint 1 is to deliver the core features of the **AI-Based Personal Finance and Budget Planning System**.

Sprint Backlog

The first sprint focuses on implementing the essential functionalities required for users to access the system and manage their finances. The selected user stories and development tasks are:

- **User Registration and Login**
 - Design the registration and login pages.
 - Implement user authentication.
 - Validate user inputs.
 - Encrypt passwords and perform login testing.

- **Income Management**
 - Create the income entry form.
 - Store income details in the database.
 - Allow users to edit and delete income records.
 - Display income history on the dashboard.

- **Expense Management**
 - Design the expense entry interface.
 - Categorize expenses.
 - Save expense details securely.

- Test expense tracking functionality.
- **Budget Planning**
 - Develop the monthly budget module.
 - Calculate available balance automatically.
 - Display budget status.
 - Verify budget calculations.
- **Reports and Dashboard**
 - Design the dashboard layout.
 - Display income and expense summaries.
 - Generate charts and financial reports.
 - Perform functional testing.

Summary

The Sprint Backlog is designed to deliver the core functionalities of the system during the first sprint. Completing these tasks provides users with a working application for managing income, expenses, and budgets while creating a strong foundation for future sprint enhancements.

6. Estimate Effort Using Story Points

Introduction

Story Points are used in Agile to estimate the effort, complexity, and time required to complete each user story. In this project, the **Fibonacci sequence (1, 2, 3, 5, 8, 13)** is used for estimation because it is a widely accepted Agile estimation technique. Higher story points indicate greater complexity, development effort, and testing requirements.

Story Point Estimation

- The **User Registration and Login** user story is assigned **5 story points** because it involves user authentication, password encryption, input validation, and security testing. The **Income Management** feature is estimated at **3 story points** since it requires creating forms, storing data, and displaying income records. **Expense Management** is assigned **5 story points** because it includes expense categorization, database operations, and tracking functionality.
- The **Budget Planning** module is estimated at **8 story points** due to its complex calculations, budget monitoring, and integration with income and expense data.
- The **AI-Based Financial Recommendations** feature receives **13 story points** because it involves AI algorithms, financial analysis, and personalized recommendation generation. The **Reports and Dashboard** module is assigned **5 story points** as it requires generating financial summaries, charts, and visual reports.

Benefits include:

- Improves sprint planning accuracy.
- Supports better workload distribution among team members.
- Helps identify complex user stories early.
- Enhances team collaboration during estimation.
- Reduces the risk of unrealistic deadlines.
- Improves project predictability and resource management.

Justification

- **3 Story Points:** Simple features with minimal development effort.
- **5 Story Points:** Moderate complexity requiring coding, database integration, and testing.
- **8 Story Points:** Complex features involving calculations and multiple modules.
- **13 Story Points:** Highly complex AI-based functionality requiring advanced logic and extensive testing.

Summary

- Using the **Fibonacci estimation technique** helps the Agile team estimate development effort more accurately and plan sprint activities effectively.
- Story points improve workload distribution, sprint planning, and timely project delivery while ensuring realistic expectations for each user story.
- Story point estimation enables the development team to manage project complexity efficiently and maintain a balanced workload throughout the sprint.
- Accurate estimation improves sprint predictability, enhances software quality, and ensures the successful completion of project objectives within the planned schedule.

7. Present the Sprint Goal and Expected Deliverables

Introduction

The **Sprint Goal** defines the primary objective that the development team aims to achieve during the first sprint. It provides a clear direction for the team and ensures that all selected user stories contribute toward delivering a functional and valuable product. The expected deliverables represent the completed features that will be available at the end of the sprint.

Sprint Goal

The goal of the first sprint is to develop the core functionalities of the **AI-Based Personal Finance and Budget Planning System**. The sprint focuses on enabling users to register and log in securely, record their income and expenses, create monthly budgets, and view a basic financial dashboard. These features provide the foundation for future AI-based recommendations and advanced financial management capabilities.

Expected Deliverables

At the end of the first sprint, the following features are expected to be completed:

- Secure **User Registration and Login** system.
- Functional **Income Management** module.
- **Expense Management** with expense categorization.
- **Monthly Budget Planning** feature.
- Basic **Financial Dashboard** displaying income, expenses, and balance.
- Secure database integration for storing financial records.
- Initial system testing and bug fixes to ensure smooth functionality.

Summary

The successful completion of the first sprint will deliver a functional prototype that allows users to manage their personal finances efficiently. These deliverables establish a strong foundation for future sprints, where advanced features such as AI-based financial recommendations, savings goal management, and bill reminder notifications will be implemented.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog	Product backlog is logically prioritized, sprint backlog is	Backlog prioritization and sprint planning are	Prioritization or sprint planning is partially complete	Backlog prioritization

Prioritization & Sprint Planning (15%)	realistic, and task breakdown is complete.	mostly appropriate with minor issues.	with limited justification.	and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25